Adoption Barriers to AI in Software Quality Assurance Processes

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Abstract

Software quality assurance (QA) is increasingly using artificial intelligence (AI) to facilitate test-case generation, documentation, automation, defect analysis, and workflow assistance. But meaningful adoption in day-to-day QA practice is uneven. Existing studies have often concentrated on technical capabilities and potential benefits, but have provided less qualitative evidence of how QA professionals experience the adoption of AI in real work and organizational settings. The research question of this thesis is: **What human and organizational factors do QA professionals perceive as influencing the adoption of AI-supported tools in software quality assurance practice?**

An exploratory qualitative design using an interview-based survey strategy was chosen to explore in depth the perceptions, experiences and work contexts of practitioners. Purposive sampling was used to recruit 11 QA professionals with different roles, levels of experience, organizational contexts, and levels of AI exposure. Data were gathered by semi-structured online interviews and analyzed through inductive thematic analysis. The analysis progressed from 61 raw codes to 25 linked codes, 15 categories, and five final themes.

The five themes were: **AI as a support mechanism rather than replacement; Conditional trust and need for verification; Task fit, domain context, and workflow integration; Organizational support, governance, and resources; and Skills, prompting, and mindset as human adoption factors.** The results suggest that adoption is a function of a combination of human, organizational and work-related conditions that are interconnected rather than technical capability alone. Participants generally saw AI as a tool that could help with productivity, test design, documentation and automation, but emphasized that professional judgment, human review and responsibility were still essential. Trust was associated with verification while usefulness was associated with task suitability, input quality, domain context and workflow integration. Other factors that influenced whether AI

could evolve beyond limited experimentation to meaningful practice were management support, approved tools, governance and security policies, budget, training, and time for learning and experimentation.

The study concludes that the adoption of AI in software QA is not only a technical problem but a socio-technical process. AI is more likely to be used in a responsible and meaningful way if it supports rather than replaces professional judgment, if its outputs are carefully checked and if the tools fit existing tasks and workflows. Adoption also depends on organizations providing appropriate support and safeguards and practitioners having the skills and mind-set to critically use AI. The results provide context-specific insights from QA professionals and are not intended to be statistically generalizable.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
QA	Quality Assurance
ROI	Return on Investment
SaaS	Software as a Service
TOE	Technology–Organization–Environment
UAT	User Acceptance Testing
UTAUT	Unified Theory of Acceptance and Use of Technology
UX	User Experience

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1

Introduction

1.1 Background

Artificial intelligence (AI) is becoming increasingly relevant in software development and software quality assurance (QA). In recent years, AI-supported tools have been proposed to help activities such as test generation, defect prediction, regression testing and test-result analysis. Tools are normally presented as ways to improve efficiency, reduce repetitive manual work and enable faster delivery of software (Awad et al., 2024; Pysmennyi et al., 2025; Trifunova et al., 2024).

Conventional automation relies on predefined scripts, structured test cases and fixed rules. On the other hand, AI-based approaches attempt to make QA more flexible and more data-driven. But, the adoption of AI in QA is not only a technical decision. These tools may affect existing workflows, the skills required of practitioners, software quality practices, and the understanding and verification of AI-generated outputs by QA professionals (Golendukhina et al., 2022; Karhu & Kasurinen, 2025; Lee & See, 2004).

QA, therefore, provides a specific context for the adoption of AI, as testing involves verification and validation activities, and test artifacts generated by AI and test-execution results might need to be carefully verified before they are put into practice (Pysmennyi et al., 2025).

Therefore, successful AI adoption in QA depends not only on tool capability, but also on how QA professionals and organizations understand, trust, support, and integrate these tools into everyday testing practice. This

provides the basis for investigating AI adoption in QA as both a technical and organizational phenomenon.

1.2 Conceptual Perspective

This study approaches the adoption of artificial intelligence in software quality assurance as a socio-technical process. Therefore, the adoption of AI in QA is shaped not only by the technical features of the tools but also by how practitioners perceive them, how the QA work is structured, and how the organizational conditions support or constrain the adoption of such tools. Accordingly, the study is positioned within empirical software engineering and AI-supported software quality assurance, with attention to the practical interaction between technology, people, and organizational setting.

In this thesis, the term AI-supported QA tools is used as a broad term for generative AI assistants, AI-enabled testing tools, and similar systems that support QA work by generation, prediction, pattern recognition, adaptive automation, or automated analysis. Adoption is understood to be more than first tool use. It discusses the introduction, acceptance, interpretation, integration into work processes and meaningful use in practice of such tools. Instead of viewing them as fixed or sequential stages of adoption, the study examines adoption at various points in the process, including early experimentation, selective use, and more established integration into QA work. The human factors are trust, skills, confidence, prompting ability, readiness to learn and attitudes to change. Organizational factors include management support, training, communication, governance and compatibility of workflow and readiness for AI-supported work.

The socio-technical perspective used in this study is supported by research that examines AI in relation to both technology and the people and work practices surrounding it. Pink et al. (2025), for example, use a socio-technical software engineering perspective to examine how trust in AI is shaped through practitioners' work practices and socio-technical relationships. This is consistent with the view taken in this thesis that AI adoption in QA cannot be understood only through the technical features of the tools. Research on broader AI-adoption also shows that adoption is dependent on organizational readiness, internal alignment, leadership support and other conditions that affect the way in which technologies are introduced and utilized in practice (Madanchian & Taherdoost, 2025; Salaheldin & Hussein, 2025; Yang et al., 2024).

1.3 Research Problem

The role of software quality assurance (QA) is increasingly affected by artificial intelligence (AI), including generative AI and AI-supported testing tools. These tools are used or proposed for tasks like defect prediction, test generation, regression support, analysis of testing data, and are often introduced as means to enhance efficiency, reduce repetitive manual work and support faster release cycles (Awad et al., 2024; Trifunova et al., 2024).

But this potential despite it, the adoption of AI in the QA practice is still uneven. AI in software testing remains largely experimental, with limited evidence of realized value and few reference implementations in the industry (Karhu & Kasurinen, 2025). Research in software engineering also suggests that generative AI is more likely to be adopted when it fits existing ways of working (Russo, 2024). In QA, having technically capable tools is therefore not enough. What matters is whether those tools can be incorporated into everyday testing activities in a useful and practical way. The problem examined in this study is not simply whether AI is being adopted, but whether it can be used in a way that fits the QA context and supports meaningful use in practice.

AI adoption in QA is also not solely a technical problem. QA AI adoption may call for changes in workflows, skills, communication, trust and organizational support. QA professionals may have to learn new ways of working and assess AI-generated outputs to determine the points at which these outputs are reliable enough to support testing decisions. Organizations must also address training, process alignment, leadership support, governance, and readiness for change if AI tools are to become part of regular testing practice (Golendukhina et al., 2022; Madanchian & Taherdoost, 2025; Yang et al., 2024).

Therefore, the emphasis in this study is on AI adoption in different QA contexts rather than in a particular industry, company size, country or development method. The aim is to understand how QA professionals experience and make sense of the adoption of AI in practice and how human and organizational factors shape the acceptance, resistance, underuse or meaningful integration of AI-supported tools in QA work.

1.4 Aim of the Study

The aim of this study is to explore how artificial intelligence (AI) is adopted within software quality assurance (QA) practices from the perspective of

QA professionals. More specifically, the study seeks to understand how human and organizational factors influence the acceptance, interpretation, and integration of AI tools in real testing environments.

The aim is not to evaluate the technical performance of AI tools, compare specific algorithms, or quantitatively assess software quality results. Instead, the study is about how AI technologies are experienced in practice and what conditions promote or hinder their meaningful use in QA work.

The study has practical and academic significance. Practically, the findings may assist QA managers, test leads, engineering leaders, training functions, and tool vendors in understanding the conditions that enable and hinder responsible AI adoption in QA practice. Academically, the study adds qualitative insight in the form of practitioner-centered understanding of AI adoption in QA as a socio-technical and organizational process.

1.5 Research Question

The problem described above leads to the following research question for this study:

What human and organizational factors do QA professionals perceive as influencing the adoption of AI-supported tools in software quality assurance practice?

Although the title of this thesis emphasizes adoption barriers, the research question uses the broader term influencing factors because human and organizational conditions may either hinder or support AI adoption depending on the context. The broader wording of the research question therefore allows the study to examine both the barriers that constrain adoption and the conditions that enable meaningful use of AI-supported tools in QA practice.

This question concerns the practical and socio-technical dimensions to the uptake of AI, not the technical performance of AI tools. It explores how QA professionals perceive, experience and interpret the adoption of AI-supported tools in their work and how organizational conditions influence this process.

Therefore, this study focuses mainly on perceptions and adoption in practice including issues such as trust, workflow integration, communication, skills and organizational support. It does not aim at comparing specific tools, or at reviewing their technical performance.

1.6 Delimitations

This study is delimited to QA professionals and related positions who are actively involved in QA activities. The focus is not on developers, data scientists, end users or other stakeholders unless they are directly involved in QA work.

The study does not objectively assess software quality outcomes, automation performance, algorithmic accuracy or the technical performance of individual AI tools. It also does not compare specific AI products, companies, industry sectors, firm sizes, countries, or software development methodologies.

Such delimitations maintain the focus of the research. The study does not intend to cover all aspects of AI in software engineering but to focus on the human and organizational dimensions of AI adoption in QA practice.

2

Literature Review

2.1 AI in Software Quality Assurance

Artificial intelligence (AI) is becoming more prevalent in software quality assurance (QA) and testing. In the literature, AI is described as a tool to enhance the speed, accuracy and flexibility of testing processes, especially in a software environment with short release cycles, complex dependencies and large volumes of data (Awad et al., 2024; Trifunova et al., 2024).

Compared with static scripts and rule-based automation, AI-driven QA tools aim to support more adaptive and data-driven testing. Reported applications include defect prediction, test-case generation, regression support and automated analysis of testing data (Awad et al., 2024; Trifunova et al., 2024). Recent work also considers large language models and AI-powered tools that can support software-testing tasks such as test generation, documentation, and practitioner support in software development and QA (Pysmennyi et al., 2025; Simaremare & Edison, 2024). Vivekananthan et al. (2025) describe an AI assistant designed to support QA practitioners through question answering, contextual recommendations, and knowledge support. Karpurapu et al. (2024) found that few-shot prompting produced fewer syntax errors than zero-shot prompting when large language models generated Behavior-driven development (BDD) acceptance tests from user stories. This suggests that providing the AI with a small number of correctly structured examples can improve the quality of generated acceptance tests.

Software QA and testing cover activities across the testing process, including requirements analysis, exploratory test analysis, test-case generation, test prioritization, defect prediction, regression testing and test execution (Pysmennyi et al., 2025; Trifunova et al., 2024). These activities include both verification and validation, and QA-focused research emphasizes that AI-generated testing artifacts and test-execution results still need to be carefully verified (Pysmennyi et al., 2025). This makes QA a specific context for AI adoption, because AI-supported outputs need to stay assessable and usable within testing and quality-assurance processes.

However, the presence of these technical capabilities does not mean that AI tools will be adopted widely or sustainably in practice. Literature suggests that AI can support speed, coverage and automation but the transition from technical potential to organizationally useful application is uncertain (Karhu & Kasurinen, 2025; Russo, 2024). This distinction is relevant to the present study because AI in QA should be perceived not only as a technical development but also as a socio-technical change concerning professional work, decision-making and organizational routines.

2.2 Adoption of AI in Software Practice and Relevance to QA

AI is often presented as beneficial, but research has shown that organizations do not simply adopt new technologies based on performance potential. UTAUT describes a number of determinants that influence technology acceptance, including effort expectancy, performance expectancy, social influence and facilitating conditions (Venkatesh et al., 2003). The adoption of AI is conditioned by technological, organizational and environmental conditions and is not simply a question of the characteristics of the technology itself as shown by studies applying the Technology–Organization–Environment (TOE) framework (Yang et al., 2024). In this thesis, UTAUT and TOE are used as supporting background for understanding adoption and for considering how broader adoption conditions may apply within QA practice, but they are not used as fixed coding frameworks or analytical categories.

Adoption in this study is understood not only as initial acceptance of AI tools but also their continued use and meaningful integration into QA work processes. This means adoption encompasses whether AI tools move from early experimentation to being part of routine testing practice at the individual and organizational level.

This perspective is supported by recent work in software engineering. Russo (2024) states that, in the initial adoption of generative AI, compatibility with current workflows might be more important than some traditional factors of technology acceptance. This is particularly true for QA, where, in addition to the tool being useful, it might be rejected if it interferes with existing testing processes, generates outputs that are hard to maintain, or needs skills the practitioners have not yet acquired.

Khojah et al. (2024) found that software engineering practitioners used generative AI more often to get suggestions on how to perform their work tasks or understand unknown topics rather than to produce ready-to-use software artifacts. This implies that such tools might be used primarily to enhance the problem solving and learning of professionals, rather than to replace the work of practitioners.

The same pattern is also found in more general research on organizational adoption of AI. Yang et al. (2024) found that AI adoption depends on technology affordances and constraints, AI readiness, innovation management, competition, and regulation, and that these factors differ across organizational contexts. Their study was not focused specifically on QA; however, it supports the view that AI adoption depends on context and organizational conditions, not only on what technology can technically offer.

Previous studies in this area also vary in method and scope. Some are conceptual, review-based, survey-based or broader software engineering rather than QA specifically (Karhu & Kasurinen, 2025; Russo, 2024; Yang et al., 2024). This suggests that the existing literature provides relevant concepts of adoption, but there is still little qualitative evidence that centers around the experiences of QA professionals themselves in their daily testing work related to AI adoption.

2.3 Human and Trust-Related Factors in AI Adoption in QA

A common topic in the literature is that barriers to adoption are not only organizational or technological, but also very human. In QA work, AI outputs typically require interpretation, validation and action by practitioners. This makes trust very relevant.

Lee and See (2004) state that trust influences the reliance on automation in situations where uncertainty and complexity are involved. Their work is important in showing that problems can occur not only when users mistrust automation but also when they overtrust it. In other words, the

proper kind of reliance depends on whether trust is tuned to the system's actual capabilities. For AI-supported QA, the question is not whether testers trust a tool, but whether they know when to accept, check or reject its outputs.

This is highly relevant to AI-based testing because many current tools are difficult to interpret fully. Karhu and Kasurinen (2025) identify lack of transparency, black-box behavior, and explainability challenges as important barriers in software-testing adoption. They also say testers lose trust when they do not understand why a tool behaves the way it does, or when commercial tools are difficult to evaluate in practice. Recent QA-oriented research also mentions explainability limitations, output inconsistencies and the possibility of false negatives in AI-assisted testing scenarios (Pysmennyi et al., 2025). Overall, these results suggest a strong association between trust in AI, interpretability, evaluability and the ability to assess whether a tool is delivering value, not just plausible outputs. Pysmennyi et al. (2025) further emphasize out that AI-supported QA has to verify both the produced artifacts and the results of test execution. Pink et al. (2025) argue that trust in AI should be understood in relation to the everyday work practices and socio-technical relations of software practitioners, rather than treated as a fixed quality of the technology.

Golendukhina et al. (2022) point out the lack of training in software-quality practices as one challenge in the development of AI-enabled systems. This indicates that the capability of AI should be combined with good knowledge of software quality and testing.

Human barriers also include skills, confidence, and readiness to work with new tools. Russo (2024) and Simaremare and Edison (2024) both indicate that adoption depends on perceptions, awareness, confidence and desire to experiment among practitioners. In practice, individuals do not adopt AI because it exists, but because they understand how it fits with their work, feel able to utilize it, and judge that its outputs are helpful enough to merit their attention and effort. This is especially true in QA, where practitioners are typically not only responsible for utilizing a tool but also for appraising its risks and choosing whether its results are reliable enough to support release-related activities.

Another human concern is the role of professional judgment. Recent QA literature suggests that AI may move human effort toward review, verification and exception management, rather than eliminating the need for humans, while perhaps reducing effort. In this way, adoption may rely on

whether practitioners regard AI as supporting their knowledge or as diminishing their influence over crucial quality decisions. This helps explain why impediments to adoption cannot be reduced to technical usability. They also involve confidence, perceived risk and the role practitioners believe they should continue to play in testing work.

2.4 Organizational and Practical Barriers to AI Adoption in QA

The literature also indicates that organizational and practical conditions influence the adoption of AI in QA. One of the most relevant syntheses for this thesis is by Karhu and Kasurinen (2025). Their secondary study classifies barriers and enablers in software testing into six categories: management, processes, human resources, tools, data, and external factors. This classification is helpful because it indicates that the problems of adoption are distributed across several levels. Management issues include finance, priorities and strategy, while process issues include policy, communication, and software development processes. Human resource problems include employee skills and attitudes. Tool and data restrictions include technical limitations, data quality, privacy, and explainability, while external concerns relate to the wider ecosystem and industry conditions.

One of the most significant outcomes in the study was that a key obstacle was the perceived lack of usefulness or produced value. Even if AI is appealing in principle, practitioners still may wonder if it really saves them time, increases reliability or just adds extra labour. Karhu and Kasurinen (2025) report ambiguity about time savings, few reference implementations, and occasions when AI-generated tests lead to increased maintenance effort. This is important for the present thesis because it implies that the barriers to adoption are often practical, not ideological: people may not reject AI because they are against innovation, but because the benefits are unclear, difficult to quantify or not worth the disruption.

Other investigations support these practical concerns. The issue of integration into existing operations is addressed as well by Trifunova et al. (2024) and Pysmennyi et al. (2025). AI tools can be used for design testing, defect analysis, regression testing, or exploratory work, but the results can be inconsistent, hard to validate, or hard to govern. The same work also identifies the issues of data quality, model interpretability, security and privacy, and the organizational transition from traditional to AI-based techniques (Pysmennyi et al., 2025; Trifunova et al., 2024). These barriers are

not only technical but also related to process design, governance and team coordination.

Broader adoption studies provide useful support here. Hradecky et al. (2022) found that the confidence in the current practices of technology, financial resources, the size of the organization, and issues of data management and protection could motivate or inhibit the readiness of the organization to adopt AI. It was discovered that technological readiness positively influenced employee capacity building and this relationship was strengthened by organizational learning (Jamil et al., 2025). It shows that organizations need to have the right infrastructure, workforce development, and continuous learning in place when introducing new technologies. Yang et al. (2024) found that the adoption of AI in organizations is influenced by AI readiness, innovation management, and regulation. Besides, the organizational work at large also reveals a link between the compatibility, complexity, privacy concerns, top-management support, and readiness with the results of AI adoption (Madanchian & Taherdoost, 2025; Salaheldin & Hussein, 2025). These studies, though not specific to QA, do support the notion that adoption is not about simply buying or experimenting with technology, but about whether organizations are ready to adopt AI in their work practice. In QA situations, this can include access to training, opportunity for experimentation, realistic expectations, internal communication and process modifications that make AI use sustainable rather than superficial.

2.5 Synthesis of the Literature and QA-Specific Research Gap

A review of the literature gives three main directions. First, AI has clear technical potential in QA. The benefits of these approaches for defect detection, test generation, regression support and automation of repetitive testing activities have been shown in prior work (Awad et al., 2024; Pysmennyi et al., 2025; Trifunova et al., 2024). Second, the actual adoption of AI software testing remains limited, experimental or uneven, with evidence often pointing to uncertainty about the real value, a lack of industry reference cases and a need for a better fit with existing work practices (Karhu & Kasurinen, 2025; Russo, 2024). Third, non-technical barriers to adoption are present. These include trust, skills, communication, process alignment, management support, data conditions and wider organizational readiness (Lee & See, 2004; Yang et al., 2024).

It is also important not to think that more adoption is always the desired outcome for every situation. Some QA professionals may limit or reject the use of AI for valid reasons like accountability, data privacy, regulatory sensitivity or the need to maintain human control of quality-critical decisions. Thus, adoption in QA may be selective and negotiated, rather than universally progressive (Karhu & Kasurinen, 2025; Lee & See, 2004).

The literature also shows a methodological imbalance. Much of the existing work is technical, conceptual, survey-based, review-based, or concerned with broader software engineering, rather than directly with QA practitioners' experiences. Studies specific to QA are still rather limited and even the most relevant recent software-testing review by Karhu and Kasurinen (2025) is a secondary study rather than direct interview research with practitioners. This indicates that there are already useful categories and theoretical perspectives in the field, but there is still not enough qualitative evidence on how QA professionals themselves understand and experience AI adoption in real work settings.

From the literature review, three interrelated levels of understanding AI adoption in QA emerged. At the individual level, trust, confidence, prompting ability, perceived usefulness and willingness to experiment influence whether practitioners feel ready to use AI-supported tools. At the organizational level, management support, governance, training, communication and workflow alignment influence the transition from experimentation to routine practice. At the task and tool level, adoption is influenced by the type of AI tool, the quality of input requirements and the fit between the tool and existing QA activities.

This thesis aims to fill this gap by addressing QA professionals' perceptions and by conceptualizing AI adoption as a socio-technical process, affected by human and organizational factors. The study does not aim to measure technical performance or to test a predefined acceptance model, but to investigate how practitioners experience adoption barriers in practice, how they view the role of AI tools in relation to their work, and what organizational contexts enable or hinder meaningful use. These levels are used as a conceptual orientation, while the empirical themes are developed inductively from the interview data.

3

Methodology

3.1 Research Design and Strategy

The study used an exploratory qualitative design to investigate the adoption of artificial intelligence (AI) in software quality assurance (QA) practice. The study is not intended to quantify patterns of adoption, or to test predefined hypotheses, but rather to understand how QA professionals view, interpret and experience the adoption of AI-supported tools in their work. For this reason a qualitative design was selected. Qualitative research is especially well suited to studies which aim to explore meanings, experiences and complex processes in context (Denscombe, 2014).

The study is exploratory because AI adoption in QA is still not well understood from the perspective of practitioners. Previous research has mainly examined technical capabilities (Awad et al., 2024; Pysmennyi et al., 2025; Trifunova et al., 2024), general technology-adoption theory (Venkatesh et al., 2003), and wider software-engineering contexts (Russo, 2024). Less attention has been given to how QA professionals themselves understand and experience AI adoption in their everyday testing work (Karhu & Kasurinen, 2025).

The study also has a socio-technical perspective. This implies that AI adoption is perceived not only as a technical issue of tools and capabilities, but also as a process that is shaped by human interpretation, organizational conditions, work practices, and practical constraints. The empirical analysis remained inductive, meaning that the themes were developed from the interview data and not imposed by a predefined theoretical framework.

The study adopted an interview-based survey strategy. Following Denscombe (2014), survey is treated as the research strategy and semi-structured interviews as the data-collection method. The term interview-based is used descriptively to indicate how the survey was implemented. Denscombe (2014) also explains that surveys can involve detailed scrutiny of a relatively small number of participants and can be associated with qualitative data and exploratory research. This strategy was selected because the research question asks what human and organizational factors QA professionals perceive as influencing the adoption of AI-supported tools in software quality assurance practice. Such a question is concerned with perceptions, experiences, interpretations, and practical work contexts rather than statistical relationships or causal measurement.

Alternative research strategies were considered but were not selected as the main approach. A case study could have provided detailed insight into one organization, but this thesis was not designed around one bounded company or project. An experimental design was also unsuitable because the study does not aim to test the performance of AI tools or measure causal effects. Design science was not selected because the thesis does not develop or evaluate a new artifact. For these reasons, an interview-based survey strategy was considered the most appropriate strategy for answering the research question.

3.2 Sampling Design

Participants were selected through purposive sampling. This sampling approach was suitable because the aim was not to produce statistically representative findings, but to obtain rich and relevant accounts from professionals who could meaningfully reflect on AI adoption in QA practice. In qualitative research, purposive sampling allows participants to be selected because they have experience or knowledge directly relevant to the research topic (Denscombe, 2014). Denscombe (2014) also notes that purposive sampling can be used to include a range of relevant people within

a small-scale study. In this study, this supported variation across QA roles, experience levels, organizational contexts, and levels of AI exposure.

The study focused on QA professionals such as QA engineers, software testers, test leads and QA managers. Participants were expected to have hands-on experience in software testing, and some exposure to AI-supported tools, AI-related initiatives, or discussions about AI in testing practice. This exposure did not have to mean a full-scale AI implementation. Participants may also have used pilots, experimented with or selectively used AI tools, or have been involved in organizational efforts to introduce such tools.

This broad but focused inclusion criterion was appropriate because the thesis investigates AI adoption rather than only mature AI implementation. If the study had included only professionals from organizations with fully established AI-based QA processes, it could have excluded important adoption experiences such as early experimentation, uncertainty, partial use, or resistance. At the same time, the study remained focused on QA professionals because they are directly involved in testing practice and can reflect on how human and organizational conditions influence the use of AI-supported tools in QA work.

Other roles such as engineering managers, product leads, or stakeholders involved in DevOps can also impact AI adoption in software development settings. However, these were not the focus of this study as the research question relates specifically to the perceptions and experiences of QA professionals.

Alternative sampling approaches were considered but not chosen as the primary sampling approach. A probability-based approach, such as random sampling, was not appropriate as the purpose was not to produce statistically representative findings or to estimate patterns within a larger population. Convenience sampling alone was also considered inadequate as selecting participants on the basis of easy accessibility may not have guaranteed that they had relevant QA experience and exposure to AI supported tools or initiatives. Hence, purposive sampling was more suitable for this study since it allowed the study to apply the defined inclusion criteria and select participants who were able to provide relevant accounts of AI adoption in QA practice.

3.3 Data Collection Design

The data collection method for this study was semi-structured one-to-one interviews. This was the appropriate method because the research question is concerned with the perceptions and experiences of QA professionals in relation to AI adoption, including issues of trust, usefulness, fit to workflow, skills, organizational support and barriers to integration. The semi-structured nature of the interviews implied that the researcher had a common interview guide for all participants, but also the freedom to ask follow-up questions, clarify responses, and probe examples specific to the role and work context of the participant.

This flexibility was important because AI adoption in QA can be experienced differently depending on the participant's role, organizational setting, level of AI exposure, and type of QA work. A fully structured interview would have limited the opportunity to explore these differences in depth, while an unstructured interview could have made it harder to compare responses across participants. Semi-structured interviews therefore provided a balance between consistency and flexibility and were suitable for collecting detailed qualitative data while keeping the discussion aligned with the research aim (Denscombe, 2014).

Other possibilities for data collection were also considered. Although a questionnaire may have gathered responses from a larger number of participants, it would have offered less opportunity to explore in detail experiences, examples and explanations behind participants' views. While observation might have illuminated how AI tools are used in practice, it would have needed access to specific organizational contexts and ongoing QA work. A document review, e.g., reviewing internal AI policies or testing guidelines, could have offered contextual information, but would not directly capture the own perceptions and experiences of QA professionals. Therefore, semi-structured interviews were the most appropriate method for the collection of data for this research.

The interview guide was organized around the research question and literature review, moving from general perceptions to concrete examples from participants' QA work.

3.4 Data Analysis Design

Thematic analysis was used to inductively analyze the interview data. This approach was appropriate because the study aimed to identify repeated

patterns of meaning across the participants' accounts, whilst remaining open to how the QA professionals themselves described AI adoption in practice. Thematic analysis is a flexible method used for identifying, analyzing and reporting patterns within data (Braun & Clarke, 2006). This made it suitable for the current study which aimed to explore perceptions, experiences and interpretations rather than test a fixed theoretical model.

The analysis was inductive in the sense that coding and theme development were guided mainly by the interview data rather than by a pre-existing coding framework. Braun and Clarke (2006) explain that inductive thematic analysis allows patterns to be developed from the data itself. At the same time, the researcher remained aware of the adoption-related concepts discussed in Chapter 2. These concepts were treated as sensitizing concepts rather than fixed codes. This means that the literature informed the researcher's general awareness, but the empirical themes were developed from the interview data rather than forced into predetermined theoretical categories.

The analysis was conducted based on Braun and Clarke (2006)'s six phases of familiarising with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes and producing the report. These phases served as a flexible guideline, not as a linear procedure. This was suitable because the analysis required repeated movement between transcripts, codes, categories, and themes as the interpretation developed.

The analysis mainly focused on semantic themes, meaning themes developed from what participants explicitly said. This level of analysis was sufficient because the study was directed at understanding practitioners' perceptions and experiences in a clearly defined and practice-oriented way, and not at developing a highly abstract interpretive theory.

Alternative data-analysis approaches were considered but were not selected. A deductive thematic analysis using predefined categories from established adoption models could have supported direct comparison with existing theoretical constructs. However, it was not selected because it could have restricted the analysis to factors identified in advance rather than remaining open to factors raised by participants (Braun & Clarke, 2006). A grounded theory approach was also not selected because the purpose of this study was to identify and interpret patterns relevant to the existing research question, rather than to develop a new theory of AI adoption (Denscombe, 2014). Inductive thematic analysis was therefore considered the most appropriate approach for this study.

3.5 Ethical Considerations in Method Design

This study involved human participants and therefore required careful attention to research ethics. Ethical considerations were taken into account when designing the study, selecting the interview method, and deciding how participants would be recruited, informed, interviewed, and represented in the thesis. The ethical approach was informed by the core research-ethics principles pertaining to informed consent, voluntary participation, confidentiality and responsible data handling (Denscombe, 2014).

The focus of the study on workplace experiences and organizational practices may have made participants uncomfortable talking about internal processes, AI initiatives or critical views of tools, workflows or management. To reduce this risk, the study was designed to focus on professional perceptions and experiences, rather than on confidential company information, trade secrets, or sensitive technical details. Therefore, the interview questions were targeted to the general experience of AI adoption in QA practice, without requiring interviewees to reveal confidential organizational information.

The choice of semi-structured one-to-one interviews also supported the ethical design of the study. Individual interviews allowed participants to speak privately about their experiences without pressure from colleagues or managers. The study was also designed to avoid identifying participants, organizations, or specific projects. Participant labels such as P1, P2, and P3 were planned from the beginning in order to protect anonymity in the analysis and presentation of findings.

3.6 Method Application

3.6.1 Sampling Application

The final sample consisted of 11 participants recruited through professional networks and online platforms such as LinkedIn. One recruitment method was a LinkedIn post inviting QA professionals, software testers, test leads, and QA managers to participate in a 30–45 minute interview about AI adoption in software QA. The post briefly explained the thesis topic, the target participant group, the voluntary and confidential nature of participation, and the requirement that participants should have QA experience and

some exposure to AI tools or AI-related discussions. The LinkedIn recruitment post and the topic overview included in the Google Meet invitations are provided in Appendix F.

Recruitment focused on professionals with relevant QA experience, as well as exposure to AI-supported tools, AI-related initiatives, or conversations around AI in QA practice. The participants came from different QA roles, experience levels and organizational contexts. This variation helped capture different perspectives on AI adoption such as practical use, cautious evaluation, organizational support, workflow fit, and barriers to implementation.

The study did not compare participants by country, organization or company type. Instead, the sample was used to capture context-sensitive accounts from QA professionals with varying roles, experience and work contexts. Therefore, the results are presented as qualitative insights into participants' perceptions of the adoption of AI in QA practice and not as statistically generalizable comparisons. Participants were anonymized using identifiers such as P1, P2 and P3. An anonymized participant overview, relevant to the interpretation of the findings, is presented in Chapter 4.

3.6.2 Data Collection Application

Data were collected through one-to-one semi-structured online interviews conducted via the video conferencing platforms Zoom or Google Meet, depending on participants' preferences and availability. The participants were in different professional and organizational settings and online interviews were more convenient and accessible for the participants.

Before starting the interviews, the participants were briefly introduced to the topics that were to be discussed, including their QA role, work context, experience with AI-supported tools, perceived usefulness, fit into the workflow, trust and verification, skills, organizational factors, barriers, facilitators, and future expectations. This helped the participants understand what the interview was about, before they agreed to take part.

Interviews lasted about 30 to 45 minutes, depending on the interviewee's experience and the level of discussion. An interview guide was used to keep the interviews comparable, while follow-up questions and probes were used when participants raised relevant issues or gave examples that needed clarification. The questions encouraged participants to describe concrete situations from their own QA work. For example, participants were asked to describe a situation where they used, considered using, or decided not to use an AI-supported tool in testing.

The interview guide was pretested for clarity, relevance, and alignment with the research question before the main data collection. It was also piloted with one person who had relevant QA experience. Minor revisions were made to improve the clarity, flow, and depth of the questions. The pilot interview was not included in the main analysis.

Participants consented to be interviewed and recorded. Recording allowed the researcher to return to the data in detail, avoid relying only on memory, and preserve the wording and meaning of participants' accounts as accurately as possible (Denscombe, 2014). The recordings were transcribed for thematic analysis. The full interview guide is included in Appendix A.

3.6.3 Data Analysis Application

After the interviews were transcribed and cleaned, the researcher familiarized himself with the data by reading through the transcripts multiple times and writing first-pass notes for each participant. These notes helped summarize important points, repeated ideas, and early observations from each interview.

Initial codes were then generated from meaningful parts of the transcripts. The codes captured ideas relevant to the research question and reflected how participants described their experiences, concerns, practices, and work contexts. Coding was carried out across the dataset so that similarities and differences between participants could be compared.

After initial coding, related codes were grouped into broader categories and candidate themes based on similarities in meaning. These groupings were then reviewed against the coded extracts and the full set of transcripts to check whether each theme was supported by the data, clearly distinguishable, and relevant to the research question. Some codes and categories were merged, refined, or reorganized during this process. The final themes were then identified and named to reflect the major patterns in participants' accounts.

Representative quotations were selected for the results chapter to show how each theme was grounded in the interview data. The quotations were chosen because they clearly illustrated the meaning of each theme while preserving participants' own wording and context. An example of how interview data were coded and developed into themes is provided in Appendix C. An example of the transcript cleaning approach used before analysis is provided in Appendix E.

3.6.4 Ethical Considerations in Method Application

Participation was voluntary during the application of the method. Participants were informed about the purpose of the study, the type of questions that would be asked, how the data would be used, and their right to withdraw before the data had been fully integrated into the analysis. Informed verbal consent was obtained at the beginning of each interview. The verbal consent script is provided in Appendix D.

Participants were also informed about confidentiality and audio recording before each interview. They were told not to reveal any confidential company information, trade secrets or sensitive technical information. With participant consent, interviews were audio recorded to support accurate transcription and analysis.

After data collection, personal identifiers were removed from the transcripts and final thesis. References to organizations, products, or projects were generalized where there was a risk of identification. Interview recordings and transcripts were stored securely, access was limited to the researcher, and the material was used only for academic purposes related to this thesis. Selected quotations were also reviewed carefully to avoid exposing participants or their organizations. After the thesis process and any required retention period are completed, the interview recordings and transcripts will be securely deleted.

The study also looked at the time and working conditions of the participants. Interviews were arranged based on participant availability, were conducted through accessible online platforms and were of a reasonable length. In summary, the ethical application of the method aimed to minimize risks, ensure confidentiality and provide a safe and respectful way for participants to share experiences.

3.7 Use of Software and AI Tools in the Research Process

Several software and AI-supported tools were used during the thesis process. The online interviews were carried out by means of video conferencing platforms like Zoom and Google Meet. Google Docs was used for transcript preparation and manual transcript cleaning. Google Sheets and Microsoft Excel were used to organize first-pass notes, codes, categories, and theme-development material. Microsoft Word was used for thesis writing, formatting, and document preparation.

AI-supported tools were also used during the revision process. Grammarly was used for grammar, spelling, and readability support. ChatGPT was used to support language refinement and reference-format checking. The university thesis feedback bot was used to receive draft-level feedback on the thesis.

4

Findings

4.1 Participant Overview and Analytical Structure

This chapter presents the results of the semi-structured interviews with 11 QA professionals about the adoption of AI-supported tools in software quality assurance practice. The purpose of the analysis was to explore how participants described the human and organizational factors that influence the acceptance, interpretation, and integration of AI-supported tools in their QA work. The findings are presented as five themes developed through inductive thematic analysis.

Different QA roles, experience levels, work contexts, and levels of AI exposure were represented among the participants. While some participants described AI as primarily experimental or selectively used, others described more mature or highly integrated AI use in QA workflows. The variation helped cover different perspectives on AI adoption while keeping the focus on QA professionals. Table 4.1 presents the participant overview.

Table 4.1: Overview of interview participants

Participant	**Role / position**	**QA experience**	**Work context / QA focus**	**AI exposure**
P1	Software QA Engineer	4+ years	Banking and business systems; mixed QA, mainly manual testing	Low–moderate
P2	Testing Trainer and Consultant	39 years	Cross-industry testing domains; coding-heavy testing practice	High
P3	Associate Tech Lead in Software QA	9 years	Healthcare product environment; automation and system-level QA	Low–moderate
P4	Quality Engineer	2.5 years	Large-scale super app / mobile platform; manual, automation, and performance testing	High
P5	Lead QA Architect	14+ years	Banking / financial systems; mixed QA, mainly manual testing	Moderate
P6	Founder / Practitioner / Researcher	11+ years	Enterprise SaaS and QA leadership; mixed manual, automation, and AI-supported QA	High
P7	QA Lead	10+ years	Real estate product / agile team; mixed manual and automation testing	Moderate
P8	Senior QA Engineer	7 years	Energy industry / data-quality testing; hybrid backend, data, and API QA	Moderate–high
P9	Technical Specialist Team Lead	10+ years	Publishing platform / microservices; fully automation-focused QA	Very high
P10	Manager of Quality Engineering	10+ years	Web, mobile, and API QA in an agile environment; mixed manual and automation testing	Moderate–high
P11	Senior Quality Leader / AI QA Advisor	25 years	Big tech, healthcare, entertainment, and AI testing platform; automation and quality leadership	High

The sample was relevant to the study because it included participants with direct experience of QA work and varying degrees of exposure to AI-supported tools or AI-related discussions. The participant group included QA engineers, leads, managers, consultants, and senior quality leaders. Their work contexts included banking, healthcare, mobile platforms, enterprise SaaS, real estate, data-quality testing, publishing platforms, web and API testing, and AI-related quality leadership. This range supported the study's aim of understanding AI adoption across different QA contexts, without treating the findings as statistically generalizable.

The analysis followed an inductive thematic-analysis process. The interview data were initially coded into 61 raw codes. The responses from participants showed that these raw codes carried more specific meanings, such as critical financial logic, healthcare/compliance, production risk, requirement clarity, prompt quality and workflow integration gaps. The raw codes were then consolidated into 25 linked codes to reduce overlap and group closely related meanings. These linked codes were further organized into 15 categories, which were finally developed into five themes. Table 4.2 presents the final theme, category, and linked-code structure used to organize the findings, while Appendix B provides the fuller traceability from raw codes to linked codes, categories, and themes.

Table 4.2: Theme, category, and code structure

Theme	**Category**	**Codes**
AI as a support mechanism rather than replacement	Productivity support	Test-case generation support; Automation/coding support; Documentation and ticketing support; Time saving / workload reduction; Coverage improvement
	Human judgment remains central	Human review required
	AI as assistant/co-pilot	Exploratory AI use
Conditional trust and need for verification	Partial/task-dependent trust	Partial trust
	Verification practices	Validation against requirements/system behaviour; Hallucination / wrong direction; Rework caused by AI
	Risk and task criticality	Risk-sensitive AI use
Task fit, domain context, and workflow integration	Structured vs complex tasks	Task-dependent usefulness
	Input and context quality	Requirement/input quality; Generic or irrelevant output
	Workflow integration level	Formal AI adoption; Deep workflow integration; Workflow integration gap
Organizational support, governance, and resources	Leadership and management support	Management support
	Governance, security, and policy	Governance and security policy
	Resources and enablement	Budget and ROI; Training and time to learn
Skills, prompting, and mindset as human adoption factors	Prompting and AI literacy	Prompting skill
	QA foundations and critical evaluation	QA fundamentals and domain knowledge
	Mindset and adaptation	Mindset shift

To keep the Results chapter focused, Table 4.2 presents the final analytical structure using the five themes, 15 categories, and 25 linked codes. A fuller code-to-theme traceability table, showing how the linked codes were

developed from the 61 raw codes, is provided in Appendix B. An example of coding and theme development using the finalized analysis structure is provided in Appendix C.

The following sections present each theme in this order. Each theme explains the categories that formed it and uses participant evidence to show how the finding relates to the human and organizational factors influencing AI adoption in QA practice. Broader interpretation and comparison with previous literature are reserved for Chapter 5.

4.2 AI as a Support Mechanism Rather Than Replacement

This theme shows that participants generally viewed AI-supported tools as useful aids in QA work, but not as replacements for QA professionals. Across the interviews, AI was described as helpful for activities such as generating test ideas, preparing test cases, supporting automation work, improving documentation, reducing workload, and giving QA professionals a faster starting point. However, participants also made clear that AI outputs still needed to be reviewed, refined, and interpreted by humans before they could be used in real QA work.

The theme brings together evidence on productivity support, continued human judgment and the use of AI as an assistant or copilot. Participants valued AI when it helped them complete routine or structured tasks more efficiently. For example, P5 described AI-supported test-case generation as useful because it reduced the time needed to prepare test cases:

> *"From this year onwards, moving to AI for test case generation has been a very time-efficient decision. We can generate test cases quickly and only need a short time to double-check what the AI tool has produced."*

This indicates that AI was perceived to be useful in improving productivity, but the participant still considered human checking as necessary. The same pattern has occurred with automation work. AI helped speed up the script development, P7 said, but the output still needed to be reviewed before it could be used.

> *"This significantly sped up the script development and reduced manual effort. However, I still reviewed and refined the generated*

code to ensure reliability and maintainability, because I needed to make sure everything was correct before it went into use."

Participants also described AI as useful for improving test coverage by identifying additional scenarios, possible edge cases, or areas that required further testing. This extended the supportive role of AI beyond generating an initial set of test cases, because it could also help QA professionals review whether relevant testing scenarios had been considered. However, participants still treated this as support rather than a substitute for their own review of test coverage.

These examples demonstrate that the participants did not object to using AI. They were willing to make use of it when it reduced effort or provided a useful starting point, but still believed that the responsibility belonged with the QA professional. This also extended to how they described AI as being a brainstorming or support tool, not an independent decision-maker. This was stated clearly by P1:

"In the end, the tool was useful mainly as a brainstorming support mechanism rather than a direct replacement for QA expertise. It saved some time during the early planning stage, but manual validation remained essential."

This theme demonstrates that the adoption of AI in QA was dependent on whether participants viewed AI as a helpful tool that could increase productivity while maintaining responsibility, validation, and professional judgment with QA practitioners.

4.3 Conditional Trust and Need for Verification

This theme shows that participants did not consider AI-generated outputs to be automatically reliable. The participants judged AI outputs according to the task, expected accuracy, available context, and possible consequences of mistakes.Therefore, trust in AI was conditional. This was particularly clear in P2's example of using AI to generate a decision table:

"The AI's table took 10 seconds to generate, but it was 40% wrong. I only know that because I spent hours constructing my own version. So that tells me: do not use AI to create a decision table that has to be correct for a given piece of code."

This quotation helps explain why participants were careful about relying on AI outputs in QA work. Although the AI produced the result quickly, the participant was able to recognise the errors only because they had enough experience to compare it with their own work. This reflects the main point of the theme: AI outputs may seem useful at first, but they still need to be checked before they can be trusted.

Other participants described this cautious position more generally. P1 explained:

> *"I would say I trust AI outputs partially, but not fully. I see them as support tools rather than sources I can directly rely on without validation."*

This shows that trust was neither complete nor absent. The AI output might be considered, but it was not seen as final evidence. This is relevant in QA work since outputs often need to match requirements, system behaviour, expected results and business rules.

Participants therefore described verification as a practical part of using AI in QA. P3 explained that AI-generated outputs would need to be checked against requirements, system behaviour, and execution results before being used:

> *"I would always compare it against actual requirements, system behaviour, and expected results. If it is related to automation, I would run and validate the script in a real environment. Basically, nothing goes into the system without manual review and execution-based validation."*

This indicates that verification was not just a general preference but part of the quality-control process. Participants expected that AI output would be compared, tested and reviewed before entering real QA work. In this sense, AI adoption created an additional trust-management step: QA professionals had to decide whether the output was accurate enough, contextually relevant, and safe to use.

The findings also showed that verification could involve additional effort when AI produced incorrect, misleading, or difficult-to-use outputs. QA professionals could need to cross-check the output against business rules and spend additional effort correcting, debugging, or maintaining it before it could be used. The level of reliance also depended on the risk of the testing context. Greater caution and verification were associated

with critical areas such as financial logic, healthcare or compliance requirements, and production systems, where an incorrect AI output could have more serious consequences. In these situations, AI use was therefore more risk-sensitive and required stronger human checking before reliance.

This theme therefore shows that trust was a key condition for AI adoption, because participants were willing to consider AI-generated outputs only when those outputs could be checked, validated, and judged safe for the task.

4.4 Task Fit, Domain Context, and Workflow Integration

This theme shows that the participants did not view AI as equally useful for all QA activities. Its usefulness depended on the task type, but also on the quality of input, domain context, and how well the tool fit into existing QA workflows. In general, AI was found to be more useful for structured or repetitive tasks such as test-case generation, support for automation, structured documentation and drafting. They were more cautious about AI usefulness when the work was complex business logic, UX judgment, defect judgment, domain-specific knowledge, unclear requirements, or close integration with existing QA processes.

The importance of input quality was clear in P5's explanation of AI-supported test-case generation:

> *"If the requirements we have received are not clear or do not contain sufficient detail, we cannot generate test cases from AI using those documents."*

This quotation shows that AI usefulness was not only shaped by the tool itself, but also by the quality of the requirements and supporting information provided to it. When the input was unclear or incomplete, participants expected AI outputs to become less useful or less reliable.

Domain context was another important condition. P8 explained that AI became more difficult to use when the work involved highly specific business logic:

> *"If you have highly domain-specific business logic, using AI becomes tricky because it does not inherently know those specific terms and scenarios. You need to feed it that context, and if you*

have not done that, or if you miss something, the outputs will be less accurate."

This suggests that the application of AI depended on the amount of contextual information available to the tool to generate relevant outputs. Domain knowledge was not seen as a minor detail but was influential to whether or not AI could support QA work in a meaningful way.

When sufficient project or domain information was not available, participants also experienced AI outputs as too generic or irrelevant to the specific testing context. This meant that contextual information was important not only for improving accuracy, but also for making the generated output sufficiently specific to the system and business area being tested.

Workflow integration also influenced how useful AI felt in practice. P7 thought AI could be useful for some development tasks but not fully integrated into the wider QA process:

"AI tools fit partially into my workflow. They integrate well during development tasks like writing automation scripts and problem solving. However, they are not fully embedded into the QA process, I use them more as support tools rather than core parts of the testing workflow, especially in structured activities like test planning, test reviews, and UAT."

This quote shows that the adoption of AI was influenced by whether AI had a place in everyday QA workflows or continued to be a separate support tool. When AI was not naturally integrated into test planning, reviews, or UAT, its use remained partial rather than central.

The level of integration was varying for participants. In some contexts AI was used experimentally and was not part of the main QA ecosystem, while in other cases it was only partially connected to normal testing activities. At the more established end, participants described formally supported AI use through approved tools or enterprise access. Some reported AI deeply embedded across their regular workflows. These differences show that workflow integration was not simply present or absent. AI adoption ranged from separate or experimental use, through partial integration, to formally supported and deeply embedded use within everyday QA work.

Therefore, this theme suggests that the adoption of AI in QA was driven by task suitability, contextual clarity and workflow fit, rather than just the general availability of AI tools.

4.5 Organizational Support, Governance, and Resources

This theme suggests that the adoption of AI in QA was not only driven by individual interest or technical usefulness. Participants also identified organizational conditions as important, including management support, security and governance policies, budget approval, tool access, and time for learning. These factors were particularly relevant as participants were working in contexts such as banking, financial systems, healthcare-related products, enterprise platforms and large-scale QA environments where tool use was often dependent on organizational approval and control.

The importance of management support for adoption was identified in several participant accounts. P2 was especially strong on this:

> *"How management supports you is absolutely key, it is the most critical factor."*

This shows the adoption of AI was not only an individual decision of QA professionals. Even when participants were interested in using AI, adoption still required the organization to support use, provide direction, and create space for AI-supported work. P9 also talked about a more advanced context of adoption where management actively supported the use of tools:

> *"Management pays for all of these tools and actively expects us to use AI."*

This suggests that organizational support could make AI use more normal within QA practice, especially when tools were officially provided rather than used informally by individuals. Management support also included quality leadership and strategic direction for AI use. This meant that adoption could be supported not only by encouraging QA professionals to experiment with AI, but also by giving clearer direction about how AI should become part of quality and testing work.

At the same time, participants described organizational control as a barrier. P5 said the resistance to wider use of AI is not from the teams, but from the security policy:

> *"So the main barrier to expanding AI tool usage is organizational security policy, not resistance from the team."*

This quote demonstrates why governance and security were central to adoption. In QA environments that deal with sensitive systems, business

data or regulated work, AI tools can not be adopted just because they're useful. They also need to follow organizational policies for data protection, approved tools and safe use. The governance concerns identified in the analysis included privacy, restrictions on sensitive data, security requirements, and compliance obligations. These conditions could limit what information QA professionals were permitted to provide to AI tools and which tools could be used in controlled testing environments.

Adoption was also affected by resources. As P10 explained, adopting advanced AI often requires formal approval because of the cost, infrastructure and return-on-investment decisions that are part of the tools:

> *"Many advanced AI tools require licences, infrastructure, or integration costs, and we have to make a business case and request a budget from the organization. Management evaluates return on investment before approving new tools, so we have to prepare proposals."*

This shows that resource availability shaped whether AI use could move beyond individual experimentation. Resource availability also included practical access to paid subscriptions and token allocations, which could affect how much AI-supported work QA professionals were able to carry out. P11 also pointed to the importance of training and time to adopt, not just access to tools:

> *"Training and guidance should be available, and people need to be given time, not just time to learn, but time to actually work with AI tools."*

This support included not only formal training, but also onboarding and sufficient experimentation time for QA professionals to become familiar with AI tools and understand how they could be used in their actual work. This theme therefore shows that AI adoption in QA depended on whether organizations provided support, safe governance, approved resources, and practical learning time for QA professionals.

4.6 Skills, Prompting, and Mindset as Human Adoption Factors

This theme shows that the adoption of AI in QA was not simply about access to AI tools. Participants described successful use of AI as requiring

the right skills, especially the ability to write effective prompts, provide enough context, evaluate AI-generated outputs, and adapt to a different way of working; AI was not perceived as an automatic producer of useful results. Instead, participants suggested that QA professionals needed to have strong QA knowledge and AI-related skills to use it responsibly.

Prompting was one of the clearest skill-related issues. P8 explained that useful AI output depended heavily on how well the user framed the prompt and supplied background information:

> *"The trickiest part is coming up with the most accurate prompt and preparing the AI model to give the right answer. You have to provide it with enough background context, about the business, the business logic, and so on, for it to give you truly accurate answers."*

This quote demonstrates that AI literacy was not just about knowing how to get access to a tool. It also meant learning how to guide the tool with clear instructions and relevant context. Without this, AI output could become too general, incomplete or unsuitable for the QA task.

P9 also described a real incident in his organization where a poorly written prompt caused AI to delete thousands of test cases after interpreting them as legacy or redundant code. This example shows that prompting skill was not only about getting better answers, but also about setting clear boundaries so that AI-supported tools did not act beyond the intended scope. Effective use also required some understanding of how the AI tool behaved and how its outputs or actions needed to be supervised, rather than treating prompting as simply entering instructions and accepting the result.

Participants also linked effective AI use to existing QA competence. P10 explained that hands-on AI experience needed to be combined with a strong testing foundation:

> *"What makes preparation easier is hands-on experience with AI tools in real scenarios, combined with a strong QA foundation, because you need that foundation to validate what AI has generated."*

This shows that the importance of QA expertise was not reduced with AI adoption. Instead, the participants saw QA knowledge as necessary for reviewing the AI output, identifying mistakes, and deciding whether the

output was suitable for use. Domain knowledge was also part of this foundation, because QA professionals needed to understand the relevant business or system context when judging whether AI-generated testing output was appropriate for use.

Mindset was another human factor in adoption. P11 described AI adoption as requiring a change in how QA professionals approach their work:

> *"It is definitely a mindset shift. You cannot keep doing things the way you always have and expect AI to simply slot in and help. You have to start thinking differently."*

Mindset was not only about willingness to change. Participants also described openness to experimentation, resistance, fear or pressure, and concerns about the possible erosion of existing skills as factors surrounding AI adoption. This shows that QA professionals could approach AI-supported work differently: some were more willing to experiment and adapt, while others had concerns about changing established ways of working or about maintaining their own professional skills as AI use increased. This means that the adoption was not just about the technical ability to use it, but also about the willingness to learn and change existing work habits.

Thus, this theme shows that AI adoption in QA was shaped by prompting and AI supervision skills, strong testing and domain knowledge, critical evaluation, and the different ways in which QA professionals responded to changing their established practices and capabilities.

5

Discussion

This chapter interprets the findings presented in Chapter 4 in relation to the research question and previous studies. The discussion examines how the five themes explain the human and organizational factors influencing the adoption of AI-supported tools in software quality assurance practice.

5.1 Interpreting AI as Support Rather Than Replacement in QA Practice

The first theme, **AI as a support mechanism rather than replacement**, showed that participants were more accepting of AI where it helped increase productivity without taking away human judgment or responsibility. This is consistent with previous studies that have shown that artificial intelligence can help reduce repetitive tasks, support test creation and increase efficiency in testing activities (Awad et al., 2024; Pysmennyi et al., 2025; Trifunova et al., 2024). However, the current study adds a practitioner-centered perspective by showing that perceived usefulness alone was not enough.

The supportive role of AI was visible across several concrete QA activities, including generating test cases and test ideas, identifying additional scenarios and possible coverage gaps, supporting automation and script development, improving QA documentation, reducing manual effort, and providing a faster starting point for testing work. These findings show that AI can support different parts of QA practice, from test design and coverage review to automation and documentation. At the same time, increased

AI involvement in these activities did not transfer responsibility to the tool. Generated test cases still had to be checked, automation code reviewed and refined, test coverage assessed, and other outputs interpreted before they could be used in practice. This is particularly meaningful in QA because the professional role involves evaluating whether testing outputs are sufficiently appropriate and reliable for the system being tested. AI could accelerate or support QA activities while professional judgment, validation, and final responsibility remained with QA practitioners which involved a division of work as of AI adoption

From a socio-technical perspective, adoption therefore depended on an acceptable division of work between AI and QA professionals. In relation to the research question, the perceived role of AI and the preservation of professional judgment were important human factors influencing adoption.

5.2 Trust, Verification, and the Limits of AI Dependence

The second theme, **Conditional trust and need for verification**, shows that participants did not trust AI unconditionally. Instead, they adjusted their trust to the task, the context they had, the accuracy they required, and the possible consequences of errors. This is in line with the notion of appropriate reliance (Lee & See, 2004), where trust should be proportional to the actual capability of the system. This aligns with identify unreliability, limited transparency and the need to verify AI-generated artifacts and execution results which are related to QA studies. (Karhu & Kasurinen, 2025; Pysmennyi et al., 2025).

The present study gives this trust issue a more specific meaning in software QA practice. For the participants, deciding whether to rely on an AI output involved activities that are already central to testing work: comparing outputs with requirements and expected results, checking them against actual system behaviour and business rules, manually reviewing the result, and, in automation work, executing generated scripts in a real environment to determine whether they behaved correctly. Trust was therefore not simply a personal feeling about whether AI was reliable. It was established through QA verification practices that provided evidence about whether an AI-generated output was suitable for use. In this sense, AI-generated outputs themselves became items that QA professionals needed to test, review, and validate before relying on them.

The findings also show that this verification requirement can affect the practical value of AI adoption. When an output is incorrect or moves in the wrong direction, QA professionals may need to correct, debug, or maintain it, creating additional rework rather than saving effort. The required level of verification also changes with testing risk. The findings also indicated greater caution where errors involved critical financial logic, healthcare or compliance requirements, or production systems. In such contexts, the consequence of accepting an incorrect output makes stronger human checking necessary. Conditional trust in QA is therefore closely connected with both the evidence produced through testing and the risk associated with the area being tested.

This study therefore extends the broader idea of appropriate reliance by showing how it is enacted in everyday QA work. AI adoption becomes more acceptable when QA professionals can subject its outputs to familiar verification activities and retain control over whether those outputs are used. Trust and the ability to verify outputs were therefore important human factors influencing adoption, while the effort required for verification and correction could also limit the practical benefit of AI-supported QA.

5.3 How Task Fit, Domain Context, and Workflow Integration Shape AI Adoption

The third theme, **Task fit, domain context, and workflow integration**, suggests that adoption depended on whether the AI was suitable for the task, contained sufficient domain context, and fit the surrounding QA workflow. Participants thought AI would be more useful in structured and repetitive tasks and less useful in testing that requires complex business logic, UX judgment, defect judgment, or deep domain knowledge. This is in line with Russo (2024), who states that the compatibility of workflow is one of the major factors for the adoption of generative AI and with Karhu and Kasurinen (2025), who have identified barriers related to processes, tools, data, and skills in software testing.

The findings give task fit a specific meaning in QA because QA work consists of activities with different levels of structure and professional judgment. Activities such as generating test cases, supporting automation, and producing structured documentation can provide AI with a relatively defined task and expected form of output. In contrast, activities involving UX judgment, interpretation of complex business logic, or defect judgment require the tester to understand how the system should behave within its

particular context. This means that the usefulness of AI cannot be judged for QA as a whole. It depends on the particular testing activity in which the tool is being used. AI may therefore be readily adopted for some parts of QA work while remaining less suitable for others.

Requirements and domain context further shape this task fit. Test-case generation, for example, depends on sufficiently clear requirements and relevant information about the system and business context. When requirements are incomplete or when the AI lacks internal or domain-specific information, the generated output can become generic, irrelevant, or less accurate for the testing situation. This is particularly important in QA because testers use requirements, expected system behaviour, business rules, and domain knowledge to determine what needs to be tested and what a correct result should look like. Providing AI with appropriate context therefore affects whether it can generate testing material that is useful for the actual system rather than producing generally plausible suggestions.

Workflow integration adds another QA-specific condition. The findings showed differences between AI being used as a separate or experimental tool and AI becoming part of established testing activities. Where AI remained outside the main QA ecosystem or was only partly integrated, participants could use it for activities such as automation scripting or problem solving while it remained less connected to test planning, test reviews, UAT, and the wider testing lifecycle. By contrast, formally supported access through approved tools or enterprise accounts, together with deeper integration across QA activities, allowed AI to become a more regular part of the workflow. AI adoption in QA concerns not only whether an AI tool can perform an individual task, but also whether it can operate within the connected set of activities through which testing work is planned, executed, reviewed, and maintained.

The present study therefore extends the understanding of QA-specific compatibility by showing that task structure, requirement quality, domain context, and the level of workflow integration work together. The availability of AI-tools was not the key driver for AI adoption but rather the practical relevance of AI to specific QA activities and the potential of AI to evolve from isolated assistance to a meaningful integration into everyday testing practice.

5.4 Organizational Conditions Shaping AI Adoption in QA

The fourth theme, **Organizational support, governance, and resources**, shows that individual willingness to use AI was not enough for meaningful adoption in QA practice. Participants identified conditions that may enable or restrict AI use including management support, quality leadership, strategic direction, governance and security requirements, budget and tool access, training, onboarding and experimentation time. This is in line with previous studies linking AI adoption to organizational readiness, leadership support, resources, governance, privacy, and compatibility of processes (Karhu & Kasurinen, 2025; Salaheldin & Hussein, 2025; Yang et al., 2024). However, the present study gives these organizational conditions a more specific meaning by showing how they affect the actual testing activities where QA professionals may use AI.

Management and quality leadership can influence whether AI moves from individual experimentation into organized QA practice. In previous themes, participants used or considered AI for the generation of test-cases, test ideas, automation work, documentation, coverage support, and verification. Organizational encouragement is not the only factor that decides on how to carry out these activities. Quality leadership and strategic direction can also shape where AI is considered appropriate within testing work, how it should be introduced into QA workflows, and how responsibility remains with QA professionals. In this sense, management support is relevant to QA not simply because employees need encouragement, but because AI use has to be positioned within existing quality and testing responsibilities.

Governance and security requirements have an especially direct effect on context-dependent QA activities. The findings showed that useful AI output can depend on clear requirements, business logic, domain information, and internal context. At the same time, participants identified privacy, sensitive data, security restrictions, and compliance as organizational concerns. This creates a practical tension for QA work: AI may need project or system information to generate relevant test cases or other useful testing outputs, while organizational policy may restrict what information can be entered into the tool or which tools can be used. Governance can therefore influence not only whether AI is permitted, but also how useful it can become for testing activities that depend on detailed system and business context. These restrictions should not automatically be interpreted

as resistance to AI, because they can also represent the controls needed to protect sensitive systems, business information, and regulated testing environments.

Resources also influence the ability to integrate AI into daily QA work. Costs mentioned by participants were for licenses, subscriptions, infrastructure, integration and token allocation, and the need to justify these in expected value and return on investment. In QA practice this matters as long-term AI use across activities such as test generation, automation support, documentation or integrated workflow assistance requires more than one-off access to a general purpose tool. If organizations do not provide the required access and resources, AI may remain limited to individual or experimental use rather than becoming part of routine testing processes. Resource decisions therefore affect the extent to which AI can move from isolated assistance into established QA workflows.

Training, onboarding, and experimentation time are also organizational conditions for AI adoption. Participants indicated that QA professionals need opportunities to learn and try AI tools before those tools can become part of their established testing practices. This may involve experimenting with AI for activities such as test-case generation or automation support and learning how to review the resulting outputs. The important organizational factor is therefore not time to perform normal QA work, but whether practitioners are given sufficient opportunity to develop practical experience with AI-supported testing.

This study suggests that, therefore, organizational support in QA is not only about access to AI. It is about creating the conditions for the use of AI in real testing work, while protecting sensitive information, maintaining quality responsibilities, providing the right tools and resources, and giving QA professionals enough time to learn how the technology fits into their work. In relation to the research question, organizational adoption was shaped by whether these conditions allowed AI to move from limited experimentation toward secure, supported, and practical use within software QA processes.

5.5 Human Readiness, Prompting, and Mindset in AI-Supported QA

The fifth theme, **Skills, prompting and mindset as human adoption factors**, shows that access to AI tools alone did not lead to their effective use in QA practice. Participants identified important human factors as prompting

ability, understanding of tool behavior, QA and domain knowledge, critical evaluation and willingness to adapt. This is in line with research that has identified skill gaps, lack of guidance, resistance to change, and limited software-quality training as barriers to AI adoption (Golendukhina et al., 2022; Karhu & Kasurinen, 2025; Madanchian & Taherdoost, 2025). However the present findings show more specifically how these human factors operate when AI is used to produce or support testing work.

Prompting has a particular meaning in QA because the instruction given to AI can directly shape a testing artifact or testing action. Participants described the need to provide business context and business logic, define the intended scope, and supervise what the tool was doing. This was especially visible in the example where a poorly written prompt caused AI to interpret thousands of test cases as legacy or redundant and delete them. Prompting in this context is therefore more than asking AI a clear question. A QA professional may need to specify what is being tested, provide relevant system or business context, define boundaries around what the tool should change, and then review whether the resulting test cases or other outputs remain within the intended testing scope. Poor prompting or weak supervision can therefore create incorrect testing material or unintended changes rather than simply producing an unhelpful answer.

The findings also explain why existing QA knowledge remains important when introducing AI. Participants did not perceive prompting skill as a replacement for testing competence. In P10, a strong QA foundation was needed to verify what AI generated, so hands-on AI experience was directly related to a strong QA foundation. In practical QA work, this means that a practitioner still needs testing fundamentals, critical thinking, and relevant domain knowledge to judge whether generated test cases, scenarios, automation-related outputs, or other testing material make sense for the system being tested. The practitioner must be able to recognise missing or incorrect logic and decide whether an output is suitable for use. This connects AI capability with an existing QA responsibility: evaluating testing evidence against requirements, business rules, system behaviour, and the expected purpose of the test. AI skills therefore become useful in QA when they are combined with, rather than substituted for, testing and domain expertise.

Mindset and adaptation also have a specific relevance to QA practice. Participants described different responses to AI, including openness to experimentation, resistance, fear or pressure, and concerns about erosion of existing skills. The issue is therefore not simply whether a QA professional

has a positive attitude toward new technology. The adoption of AI may mean that practitioners need to change how they prepare testing work, interact with tools and review AI-supported outputs while continuing to use their own testing judgment. At the same time, concerns about skill erosion are particularly relevant where practitioners still need enough independent QA knowledge to recognise when AI-generated testing output is incomplete or incorrect. From this perspective, openness to AI needs to coexist with maintaining the testing capabilities required to supervise and verify its use.

The present study therefore suggests that human readiness for AI-supported QA is not a separate set of generic AI skills. It involves combining the ability to instruct and supervise AI with testing fundamentals, domain understanding, critical evaluation, and an adaptive but still critical approach to changing testing practices. In relation to the research question, adoption was influenced by whether QA professionals could use AI within real testing activities while retaining the knowledge and judgment needed to evaluate what the tool produced.

5.6 Overall Discussion and Implications

Together, the five themes show that the adoption of AI in software QA is influenced by interacting human and organizational conditions including professional judgment, trust, task and workflow fit, governance, resources, skills and mindset. The findings suggest that QA professionals were more open to AI when it supported concrete testing activities such as test-case generation, coverage review, automation, and documentation without removing their responsibility for reviewing and validating the output. Trust was linked to the ability to verify AI generated outputs against requirements, business rules, expected system behavior and execution results, while usefulness was dependent on the type of testing activity, the quality of the available context and the degree of fit of AI to existing QA workflows.

The findings answer the research question by showing that human factors included conditional trust, critical judgment, QA and domain knowledge, prompting and supervision ability, and willingness to adapt. Organizational factors included management and quality leadership, approval of tools, security and governance arrangements, budget and access to resources, training, onboarding and experimentation opportunities. These factors were connected to each other rather than functioning independently.

For example, trust was strengthened if verification was possible, but effective verification relied on sufficient testing and knowledge of the domain. Similarly, useful AI output depended on relevant requirements and business context, and prompting and supervision skills affected how that context was provided to the tool. The integration into the workflow was also affected by whether organizations provided approved access, resources and governance conditions that enabled AI to become part of regular testing work.

The study therefore provides practitioner-centered evidence of how AI-adoption conditions operate within the professional responsibilities of software QA. Not only does it contribute the fact that trust, skills, governance or management support matter, but also that they impact whether AI can be used responsibly in activities where QA professionals still need to judge the scope of testing, validate outputs, compare results with requirements and system behavior, and remain responsible for their quality decisions. In practice this means that organizations should identify QA activities where AI is suitable, ensure that the necessary testing context can be provided safely, retain independent review and verification, and provide the tools, resources and learning opportunities needed for practitioners to use AI within established testing workflows. This integrated account may also support future research examining these relationships in larger samples or more specific QA contexts.

The combined findings also have ethical implications. When AI is introduced into testing work, it is important to have accountability, protection for sensitive information and safeguards against over-reliance. These implications arise from the study's findings on continued human responsibility, verification, governance, and concerns about maintaining professional QA skills rather than from a transfer of quality responsibility to AI.

The societal implications of the study are indirect as the research is about professional QA practice rather than end-user outcomes. However the findings suggest that the continued human accountability, verification and protection of sensitive information are relevant when AI is used in quality-assurance processes for software that organizations and users rely on.

5.7 Methodological Reflections and Limitations

The quality of this qualitative study can be assessed using the concepts of credibility, dependability, transferability and confirmability. These are

commonly used to discuss trustworthiness in qualitative research and thematic analysis (Denscombe, 2014; Nowell et al., 2017). Denscombe (2014) explains that qualitative research cannot normally be assessed through exact replication in the same way as experimental research because social settings change and the researcher is closely involved in collecting and interpreting the data. Instead, the quality of qualitative research depends on showing that the data and interpretations are credible, that the research process is sufficiently transparent, that the possible transfer of findings can be judged from the study context, and that the influence of the researcher on the analysis is acknowledged.

Credibility assesses whether the data and interpretations are reasonably accurate and appropriate (Denscombe, 2014). In this study, semi-structured interviews were used as a method for supporting the credibility as this method allows the participants to explain their experiences and follow-up and clarification could be made when needed. The interviews were recorded with consent and transcribed for analysis, and the thematic analysis involved returning to the transcripts when reviewing codes, categories, and themes. The findings were therefore grounded in participants' interview accounts, and representative quotations were used in Chapter 4 to show how the themes related to the underlying data. No respondent validation or member checking was conducted, however, meaning that the participants were not asked to review the final interpretations. No triangulation with other data sources or independent researchers was conducted. Also, the study is based on the reports of participants' experiences and perceptions, not on direct observation of their use of AI in QA practice. These limitations mean that the findings should be interpreted as an account of how the participating QA professionals perceived and described AI adoption rather than as independent verification of their actual AI-use behaviour. Denscombe (2014) identifies grounded empirical data, respondent validation, and triangulation as possible ways of strengthening credibility in qualitative research.

Dependability concerns whether the research process and the decisions made during it are sufficiently clear for others to understand and evaluate. Denscombe (2014) argues that, because exact replication is difficult in qualitative research, researchers should provide a transparent account of their methods, analysis, and decision-making. In this study, the sampling criteria, interview procedure, data-collection process, and thematic-analysis procedure were documented in Chapter 3. The analysis was also recorded from the initial coding stage through the development of linked

codes, categories, and final themes, resulting in 61 raw codes, 25 linked codes, 15 categories, and five themes. The appendices provide additional traceability between these analytical levels. This documentation allows the reader to follow how the analysis developed, although no formal independent audit of the analysis was conducted. A further limitation is that the coding and thematic interpretation were undertaken by one researcher. Because qualitative analysis involves interpretation, another researcher could potentially make somewhat different coding or grouping decisions even when working with the same interview material. Therefore, the study does not claim inter-coder reliability or researcher triangulation. Denscombe (2014) describes such transparency and documentation as important for making the qualitative research process open to evaluation.

Transferability concerns the extent to which readers can judge whether findings from a small qualitative study may be relevant to other comparable settings. Denscombe (2014) distinguishes this from statistical generalization and states that researchers should give sufficient information on participants and the context of the research so that the reader can judge the possible relevance elsewhere. The present study involved 11 purposively selected QA professionals with variation in roles, experience levels, organizational contexts, work domains, and levels of exposure to AI-supported tools. This variation provided perspectives from different QA settings rather than from a single organization or type of practitioner. However, the sample is small and was not designed to be statistically representative of the wider population of QA professionals. The findings therefore cannot be generalized statistically to all QA professionals, organizations, industries, or countries. Instead, the participant and work-context descriptions provided in the thesis allow readers to judge whether the findings may be relevant to comparable software QA environments. This limitation means that the conclusions should remain context-specific and should not be interpreted as universal conditions for AI adoption in QA.

Confirmability concerns the extent to which the findings can be connected to the research data rather than being presented only as the researcher's personal views. Denscombe (2014) suggests that it is not easy to achieve complete objectivity in qualitative research as the researcher is bound to interpret. Researcher influence during data collection was also partly limited through the use of a common semi-structured interview guide for all participants. The guide was organized around the research question and literature review and covered the same main topic areas across the interviews, while still allowing follow-up questions for clarification and

relevant examples. This helped reduce the extent to which the interview questioning depended only on issues that arose spontaneously from the researcher's own assumptions. However, the semi-structured format still required researcher judgment when deciding when and how to probe participants' responses, so this measure reduced rather than removed the possibility of researcher influence. In this study, confirmability was supported by maintaining traceability from interview data to raw codes, linked codes, categories, and final themes, and by presenting participant quotations alongside the findings. This makes it possible for the reader to see how the interpretations relate to the empirical material. At the same time, the analysis remained interpretive and was conducted by a single researcher, so the researcher's judgment influenced the coding, grouping, and interpretation of the data. No independent second researcher was used to confirm the coding, and no formal researcher triangulation was performed. These limitations mean that the findings should be understood as a transparent and evidence-based interpretation of the interview material rather than as the only possible interpretation of the data. Denscombe (2014) also notes that qualitative researchers should be open to ambiguity, contradictions and alternative explanations, rather than accepting one interpretation as automatically definitive.

The qualitative approach provided important strengths to this study. Semi-structured interviews produced detailed accounts of how QA professionals understood and experienced AI-supported tools in their work, allowing the analysis to consider differences in trust, task suitability, organizational support, workflow integration, and professional judgment. Denscombe (2014) identifies richness of detail and grounding in real-world evidence as important advantages of qualitative analysis. At the same time, qualitative analysis has limitations, including difficulty in generalizing from small samples, the influence of researcher interpretation, and the possibility of taking statements out of their original context. These limitations were relevant to the present study and were addressed through contextual descriptions, traceability of the analysis, and careful use of participant quotations, but they cannot be completely removed.

Software and AI-supported tools were also used during parts of the research process for activities such as transcript preparation and organization, language refinement, and organizing the reference list. These tools

were used as support rather than as substitutes for the researcher's analytical responsibility. Decisions about coding, the development of categories and themes, interpretation of the findings, and the final conclusions remained the responsibility of the researcher. This distinction is important because the credibility and confirmability of the study depend on maintaining a clear connection between the original interview material, the documented analysis, and the conclusions presented in the thesis.

Overall, these methodological limitations do not invalidate the findings, but they define boundaries in which the conclusions should be interpreted. The study provides context-specific evidence of how 11 QA professionals perceived human, organizational and work-related factors that influence AI adoption. The findings should therefore be understood as qualitative insights into software QA practice that may be relevant to comparable settings, rather than as statistically generalizable claims about all QA professionals or organizations.

6

Conclusion

6.1 Answer to the Research Question

This study explored how QA professionals perceive human and organizational factors influencing the adoption of AI-supported tools in software quality assurance practice. An exploratory qualitative design using an interview-based survey strategy was used to examine the perceptions and experiences of 11 QA professionals with different roles, work contexts, and levels of AI exposure. Participants were selected through purposive sampling, data were collected through semi-structured online interviews, and the data were analyzed using inductive thematic analysis.

The research question was:

What human and organizational factors do QA professionals perceive as influencing the adoption of AI-supported tools in software quality assurance practice?

The findings suggest that AI adoption in software QA is shaped by a combination of human, organizational, and work-related conditions rather than by technical capability alone. Important human factors included professional judgment, conditional trust, the ability to verify AI-generated outputs, QA and domain knowledge, prompting and supervision ability, critical evaluation, and willingness to adapt to changing working practices. Participants in general viewed AI as a support mechanism and not a replacement of QA expertise. This was important as AI could help with things like generating test cases, reviewing coverage, automation and documentation,

but QA professionals still had to decide if the testing material produced was appropriate to use. Trust was therefore closely connected to the ability to verify outputs against requirements, business rules, expected system behavior, manual review or execution results.

Organizational factors included management and quality leadership, approved tools, governance and security policies, budget and access to resources, training, onboarding, and opportunities for experimentation. These conditions influenced whether AI could move beyond individual or occasional use and become part of established QA practice. Governance and security arrangements could affect which AI tools could be used and what requirements, business information, system context, or sensitive data QA professionals were permitted to provide to them. Resources and organizational support also influenced whether practitioners had suitable access to AI tools and opportunities to develop practical experience using them within testing work.

The findings also showed that these human and organizational factors interacted with work-related conditions such as task suitability, requirement and input quality, domain context, and workflow integration. AI was perceived as more useful for some structured or repetitive testing activities, while activities involving complex business logic, UX judgment, or defect judgment required stronger professional interpretation. Useful AI-supported testing also depended on having enough relevant context and on whether the tool fitted existing QA processes such as automation work, test planning, test reviews, UAT, and other parts of the testing lifecycle.

Those were interconnected conditions. Trust in AI was partly dependent on the verifiability of its outputs. Effective verification was based by sufficient QA knowledge, domain understanding and critical judgment. The usefulness of AI-generated testing material depended on appropriate requirements and business context, while prompting and supervision skills influenced how that context was communicated to the tool. Similarly, integration into regular QA work could depend on organizational approval, governance, resources, and access to suitable tools. AI adoption in QA was therefore not simply a matter of whether an individual practitioner was willing to use AI.

The main conclusion is that AI adoption in software QA is a socio-technical process in which human capability, organizational conditions, and the characteristics of the testing work interact. Meaningful adoption is more likely when AI supports concrete QA activities without removing professional responsibility, when its outputs can be independently verified, when the task

and available context are appropriate, when the tool can be integrated into established testing workflows, and when organizations provide safe and supported conditions for use. Therefore, adoption of AI in QA is not simply about accepting a new technology, but whether AI is able to be incorporated into testing practice, while maintaining the professional judgment, verification, domain understanding and responsibility required for software quality assurance.

6.2 Contribution, Recommendations, and Future Research

This study provides practitioner-oriented evidence on how human, organizational, and work-related conditions interact in the adoption of AI-supported tools within the framework of QA. The findings do not identify a single dominant barrier, rather it seems that meaningful adoption is dependent upon the combined influence of professional judgment, trust and verification, task and workflow fit, organizational support and governance, and practitioner skills and mindset. This integrated understanding provides a basis for further research on the adoption of AI in software QA practice.

The findings suggest that organizations and QA teams are suggested to start by identifying clearly defined QA use cases where AI offers practical value and select tasks that align with current AI capabilities. AI should be a support tool, not a replacement for QA professionals. Human review, verification and accountability must remain at the center of this, particularly in quality-critical or risk sensitive work. Outputs should be checked against requirements, business logic, system behaviour and execution results. Organizations should provide approved and secure tools together with clear governance, privacy, security, and compliance arrangements. Budgets, licences, subscriptions and expected value should also be considered in adoption plans. Training, practical guidance, and sufficient time for experimentation are needed to develop prompting, critical-evaluation, QA, and domain capabilities.

Future research with larger and more diverse samples may examine these factors. Including QA professionals, engineering managers, product roles, and security or governance specialists in multi-stakeholder studies could provide a broader organizational perspective. Longitudinal case studies that combine interviews with observations or records of tool use could show how adoption varies over time. Further research may focus

on specific QA contexts in isolation such as banking and finance, healthcare, enterprise software, highly automated QA teams and organizations with limited AI adoption. Comparative studies could further investigate variations in adoption between more and less regulated domains, and the effects of long-term AI use on the development of QA expertise.

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Appendix A - Interview Guide

Interview Guide: Adoption Barriers to AI in Software Quality Assurance Practice

Purpose of the interview

This interview is part of a master's thesis at Stockholm University. The study explores how AI tools are adopted in software quality assurance practice, with particular focus on human and organizational factors as perceived by QA professionals.

A. Opening and Consent

The standard verbal consent script used before the interview questions is provided in Appendix D.

Background Questions

1. Can you briefly describe your current role and your experience in software testing or quality assurance?
2. Can you briefly describe your work context, such as the kind of organization, team, or systems you work with?
3. Do you work mainly in manual testing, automation testing, or a mix of both?
4. Do you work in an agile, DevOps, or another development environment?

Exposure to AI-Supported Tools

5. What kind of exposure have you had to AI-supported tools in software testing or quality assurance work?
6. Were these tools introduced formally by your organization, or did you begin using them more informally?
7. What kinds of AI tools have you used, considered using, or discussed in your work?
8. For what kinds of activities are these tools used, such as test case generation, documentation, requirement understanding, reporting, debugging, or automation?

Specific Experience with AI Tools

9. Can you describe one specific situation where you used, considered using, or decided not to use an AI-supported tool in testing work?
10. What happened in that situation, and what influenced your decision?

Perceived Usefulness and Fit

11. From your experience, what kind of value can AI tools provide in QA work?
12. In which areas do they seem less useful?
13. Do AI tools save time in your QA activities? Why or why not?
14. Have AI tools ever created extra work, uncertainty, or rework?
15. How well do AI-supported tools fit into your existing QA workflow and everyday testing practice?
16. Are there any practical integration difficulties when using these tools?

Trust and Evaluation

17. How do you feel about relying on AI-generated outputs in testing work?
18. Do you trust AI outputs fully, partially, or only a little?

19. Are there some tasks where you trust AI more than others?
20. When AI gives you a suggestion or output, how do you usually verify or evaluate it?

Readiness and Learning

21. How prepared do you feel to use AI-supported tools in your QA work?
22. What makes it easier or harder to be prepared?
23. Have you faced any learning difficulties or uncertainty when trying to use or understand AI tools in testing?
24. Are there any knowledge or skill gaps that affect adoption in your team or organization?

Organizational Factors

25. What organizational factors influence whether AI tools are adopted in QA work?
26. Does management support influence AI adoption in your context?
27. Are training, guidance, or learning opportunities available for AI-related work?
28. Are there any concerns around privacy, security, governance, or accountability when using AI tools?
29. In your experience, what makes it difficult for AI tools to move beyond experimentation into regular use?

Barriers and Future Outlook

30. Overall, what do you think are the main barriers to AI adoption in software QA practice?
31. What factors make adoption easier or more likely to succeed?
32. Looking ahead, what would need to happen for AI adoption in QA to become more meaningful or successful in practice?

Closing Questions

33. Is there anything important about AI adoption in software QA that we have not yet discussed?

34. Is there anything else from your own experience that you would like to add before we finish?

Closing Script

Thank you very much for your time and for sharing your experience. Your input is very valuable for this study.

Appendix B - Code-to-Theme Traceability Table

This appendix provides the code-to-theme traceability table used in the thematic analysis. The table shows how the 61 raw codes were consolidated into 25 linked codes, which were then organized into 15 categories and five final themes. This appendix supports transparency in the analysis process, while Chapter 4 presents the final analytical structure in a shorter form.

Table B.1: Code-to-theme traceability from raw codes to final themes

Theme	Category	Linked code	Raw codes
AI as a support mechanism rather than replacement	Productivity support	Test-case generation support	Test ideas
		Automation/coding support	Automation support
		Documentation and ticketing support	Documentation support
		Time saving / workload reduction	Faster starting point; Productivity expectations
		Coverage improvement	Coverage improvement
	Human judgment remains central	Human review required	Final decision by QA; Human review required
	AI as assistant/co-pilot	Exploratory AI use	Brainstorming; Co-pilot; Not autonomous decision-maker; Optional support; Support assistant
Conditional trust and need for verification	Partial/task-dependent trust	Partial trust	50/50 trust; High trust only after validation; Partial trust; Trust after review
	Verification practices	Validation against requirements/system behaviour	Compare with requirements; Execution validation; Manual review
		Hallucination / wrong direction	Cross-check business rules
		Rework caused by AI	Ability to debug/maintain AI output
	Risk and task criticality	Risk-sensitive AI use	Critical financial logic; Healthcare/compliance; Production risk
Task fit, domain context, and workflow integration	Structured vs complex tasks	Task-dependent usefulness	Weaker for UX, business logic, defect judgment; Works for repetitive/structured tasks
	Input and context quality	Requirement/input quality	Internal context; Requirement clarity
		Generic or irrelevant output	Domain information
	Workflow integration level	Formal AI adoption	Approved tools; Enterprise accounts
		Deep workflow integration	Deeply embedded workflow
		Workflow integration gap	Outside ecosystem; Partially integrated
Organizational support, governance, and resources	Leadership and management support	Management support	Management encouragement; Quality leadership; Strategic direction
	Governance, security, and policy	Governance and security policy	Compliance; Privacy; Security restrictions; Sensitive data
	Resources and enablement	Budget and ROI	Budget; Subscriptions; Token allocation
		Training and time to learn	Experimentation time; Onboarding; Training
Skills, prompting, and mindset as human adoption factors	Prompting and AI literacy	Prompting skill	AI supervision skills; Context design; Prompt quality; Tool understanding
	QA foundations and critical evaluation	QA fundamentals and domain knowledge	Critical thinking; Domain knowledge; Manual verification; Testing fundamentals
	Mindset and adaptation	Mindset shift	Fear/pressure; Mindset change; Openness to experimentation; Resistance; Skill erosion concerns

Appendix C - Example of Coding and Theme Development

Table C.1 provides five illustrative examples of how selected interview excerpts were connected to raw codes, finalized linked codes, categories, and themes. The examples follow the finalized analytical hierarchy used in this study. The complete traceability across 61 raw codes, 25 linked codes, 15 categories, and five themes is presented in Appendix B.

Table C.1: Example of coding and theme development

Participant	**Interview excerpt**	**Raw code**	**Finalized linked code**	**Category**	**Theme**
P4	"Human intervention is always required at the end of the day, and yes, we explicitly have to review what the AI is doing."	Human review required	Human review required	Human judgment remains central	AI as a support mechanism rather than replacement
P1	"I would say I trust AI outputs partially, but not fully. I see them as support tools rather than sources I can directly rely on without validation."	Partial trust	Partial trust	Partial/task-dependent trust	Conditional trust and need for verification
P5	"If the requirements we have received are not clear or do not contain sufficient detail, we cannot generate test cases from AI using those documents."	Requirement clarity	Requirement/ input quality	Input and context quality	Task fit, domain context, and workflow integration
P5	"So the main barrier to expanding AI tool usage is organizational security policy — not resistance from the team."	Security restrictions	Governance and security policy	Governance, security, and policy	Organizational support, governance, and resources
P11	"It is definitely a mindset shift. You cannot keep doing things the way you always have and expect AI to simply slot in and help. You have to start thinking differently."	Mindset change	Mindset shift	Mindset and adaptation	Skills, prompting, and mindset as human adoption factors

Appendix D - Verbal Consent Script Used in the Interviews

Verbal Consent Script Used in the Interviews

At the beginning of each interview, the following verbal consent script was read to the participant:

Hello, and thank you for taking the time to participate in this interview.

My name is Haritha Subasinghe, and I am conducting this interview as part of my master's thesis at Stockholm University. The study explores how AI tools are adopted in software quality assurance practice, especially from the perspective of QA professionals.

Before we begin, I would like to briefly explain a few important points:

- Your participation is voluntary.
- You may choose not to answer any question.
- You may stop the interview at any time.
- This interview will be used only for academic research purposes related to this thesis.
- You may also withdraw from the study at any time before your interview data have been fully integrated into the analysis.
- Your identity and organization will be treated confidentially in the final thesis.
- With your permission, I would like to audio record this interview to support accurate transcription and analysis.

Do you agree to participate in this interview and to the audio recording?

Note

Informed consent was obtained verbally at the beginning of each interview before the questioning started.

Appendix E - Example of Transcript Cleaning Approach

Example of Transcript Cleaning Approach

The interview recordings were transcribed and the initial transcripts were manually cleaned before thematic analysis. The cleaning process aimed to improve readability while preserving the participants' intended meaning. Corrections focused on clear transcription errors, repeated or filler wording, grammar, technical terms, punctuation, and sentence flow. No new ideas were added during this process.

Table E.1 presents examples from the initial and cleaned transcripts. The examples show that cleaning was applied only where necessary. Statements that were already clear were retained without unnecessary changes.

Table E.1: Examples of transcript cleaning

Participant	**Initial transcript excerpt**	**Cleaned transcript excerpt**	**Cleaning performed**
P8	"So, as one part of my duty is to make sure the data quality in these data leak houses and all."	"One key part of my role is ensuring data quality in these data lakehouses."	Corrected the technical term "data leak houses" to "data lakehouses" and improved grammar and sentence flow.
P8	"The difficulty or the trickiest part is or is kind of like using the most accurate prone or coming up with the most accurate prompt and also preparing the AI model to get the right answer."	"The trickiest part is coming up with the most accurate prompt and preparing the AI model to give the right answer."	Removed repeated and filler wording, corrected the transcription error "prone" to "prompt," and improved sentence flow without changing the meaning.
P1	"I would say I trust AI outputs partially, but not fully."	"I would say I trust AI outputs partially, but not fully."	No change was needed because the statement was already clear.

Appendix F - Recruitment Materials

LinkedIn recruitment post

The following recruitment message was posted on LinkedIn:

I'm currently doing my Master's thesis at Stockholm University on AI adoption in Software QA.
I'm looking for QA professionals, testers, test leads, or QA managers for a short 30–45 minute interview.
The study focuses on how AI is used or discussed in QA, especially the human and organizational factors that influence adoption.
Participation is voluntary and confidential.
If you have experience in QA and some exposure to AI tools or AI-related discussions in QA, please message me directly.
Thank you for your support.

Google Meet invitation note

The following topic overview was included in the Google Meet invitation:

Interview topics:

- QA role and work context
- Exposure to AI tools in QA
- Practical experience with AI-supported tools
- Perceived usefulness and workflow fit
- Trust, verification, and human judgment
- Skills, readiness, and learning challenges
- Organizational factors affecting AI adoption
- Main barriers, facilitators, and future outlook

This interview is part of a master's thesis at Stockholm University. Participation is voluntary, and responses will be treated confidentially.